The t-table

The t-table given is the probability integral of t-distribution. It gives over a range of values of v at different levels of significance. By selecting a particular degrees of freedom and level of significance, we determine the tabular value of t. We establish a null hypothesis, and if our computed t is greater than the tabular t, we reject the null hypothesis; if our computed tis smaller than the tabular t, we accept the null hypothesis.

Applications of t-distribution

The following are some important applications of the t-distribution :

(1) Test of Hypothesis about the population mean.

(2) Test of Hypothesis about the difference between two means with independent samples.

(3) Test of hypothesis about the difference between two means with dependent samples.

(4) Test of hypothesis about coefficient of correlation.

(l) Test of Hypothesis about the Population Mean (σunknown and sample size is small)

When the population distribution is normal and standard deviation σis unknown then the "t" statistic is defined as:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

follows the Student's t-distribution with *(n - l) d.f.* where:

$\bar{x}$ = sample mean, μ = hypothesized population mean and n= sample size

and s is the standard deviation of the sample calculated by the formula:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

The null hypothesis to be tested is whether there is a significant difference between $\bar{x}$ and μ.

If the calculated value of *t* exceeds the table value of *t* at a specified level of significance, the null hypothesis is rejected and the difference between $\bar{x}$ and μ is regarded significant. If the calculated value of tis less than the table value, the difference between $\bar{x}$ and μ is not considered to be significant. It may be noted that this test is based on

n - 1 degrees of freedom.

Confidence Interval for the Population Mean

When sampling is from a normally distributed population with unknown σ,the 100 (1 - α) per cent confidence interval for the population is given by:

$$\bar{x} - t_{\alpha/2}(n-1)s/\sqrt{n} < \mu < \bar{x} + t_{\alpha/2}(n-1)s/\sqrt{n}$$

Example 1

Ten oil tins are taken at random from an automatic filling machine. The mean weight of the tins is 15.8 kg and standard deviation is 0.50 kg.Does the sample mean differ significantly from the intended weight of 16 kg?

Solution

Let the null hypothesis be that the sample mean weight is not different from the intended weight.

Given that $n = 10,\ \bar{x} = 15.8,\ s = 0.50,\ \mu = 16$

Using the t-test, we have

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{15.8 - 16}{0.5/\sqrt{10}} = \frac{-0.2}{0.158} = -1.266$$

The table value of t for 9 d.f. at 5% level of significance is 2.26. The computed value of t is smaller than the table value of t. Therefore, the difference is insignificant and the null hypothesis is accepted. Hence the difference between sample mean weight
and the intended weight is insignificant.

Example 2

Prices of shares (in $) of a company on the different days in a month were found to be:

66, 65, 69, 70,69, 71,70, 63, 64 and 68.

Test, whether the mean price of the shares in the month is 65.

Solution

We want to test the hypothesis:

$H_0 : \mu = 65$ *versus*
$H_0 : \mu \neq 65$

To test the above hypothesis, we compute;

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

From our calculator, $n = 10,\ \bar{x} = 67.5,\ s = 2.8,\ \mu = 65$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{67.5 - 65}{2.8/\sqrt{10}} = \frac{2.5}{0.89} = 2.81$$

The table value t for 9 degrees of freedom at 5% level of significance is 2.26. Since the computed value of t = 2.81 is greater than the table value, we reject the null hypothesis. Hence, the mean price of the shares in the month is not 65.

SPSS OUTPUT

```
T-TEST
  /TESTVAL=65
  /MISSING=ANALYSIS
  /VARIABLES=price
  /CRITERIA=CI(.95).
```

T-Test

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
price of shares	10	67.50	2.799	.885

One-Sample Test

	Test Value = 65					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper

price of shares	2.825	9	.020	2.500	.50	4.50

Conclusion

Since computed P-value = 0.02 < 0.05, we reject H_0 and conclude that the mean price of the shares in the month is not 65.

(2) Test of Hypothesis about the Difference between Two Means

In testing a hypothesis concerning the difference between the means of two normally distributed populations when the population variances are unknown, the *t-test* can be used in two types of cases;

(a) the case in which variances are equal, *i.e.*, $\sigma_1^2 = \sigma_2^2$, *(b)* the case in which variances are not equal, *i.e.*, $\sigma_1^2 \neq \sigma_2^2$.

(a) Case of equal variances

Let the null hypothesis be that there is no significant difference between the means of the two populations, *i.e.*, $H_0 : \mu_1 = \mu_2$. When the population variances (though unknown) are equal then the appropriate test statistic to be used is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

will follow t-distribution with $(n_1 + n_2 - 2)$ d.f., where $\bar{x}_1$ *and* $\bar{x}_2$ are sample means of sample 1 of size n_1 and sample 2 of size n_2 respectively; μ_1 and μ_2 are the population means, and s is "pooled" estimate of the common population standard deviation obtained by pooling the data from both the samples as given below:

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \text{ where } s_1^2 = \frac{\sum(x_1 - \bar{x}_1)^2}{n_1 - 1}; \text{ and } s_2^2 = \frac{\sum(x_2 - \bar{x}_2)^2}{n_2 - 1}$$

If the computed value of *t* is less than the table value of *t* at a specified level of significance,the null hypothesis is accepted and the difference between the two means is regarded asinsignificant. If the computed value of *t* is more than the table value of *t,* the null hypothesis is rejected and the difference between the sample means is regarded assignificant.

***b)* Case of unequal variances**

When the population variances are not equal, *i.e.* $\sigma_1^2 \neq \sigma_2^2$,we use the unbiased estimators s_1^2 and s_2^2 to replace σ_1^2 *and* σ_2^2 .In this case, the difficulty arisesbecausethe sampling distribution has large variability than the population variability.The statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s\sqrt{\frac{s_1^2}{n_1} + \frac{s_1^2}{n_2}}}$$

maynot strictly follow t-distribution but may be approximated byt-distribution with a modified value for the degrees of freedom given by

$$d.f. = \frac{\left[s_1^2/n_1 + s_2^2/n_2\right]^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_1^2/n_1\right)^2}{n_2 - 1}}$$

Example 3

Two different types of drugs A and B were tried on certain patients for increasing weight, 5 persons were given drug A and 7 persons were given drug B. The increase in weight (in pounds) is given below:

Drug A	8	12	13	9	3		
Drug B	10	8	12	15	6	8	11

Do the two drugs differ significantlywith regard to their effect in increasing weight?

Solution

Null hypothesis $H_0 : \mu_1 = \mu_2$, there is no significant difference in the efficacyof the two drugs. Thus

We want to test the hypothesis:

$H_0 : \mu_1 = \mu_2 \quad versus$
$H_0 : \mu_1 \neq \mu_2$

To test the above hypothesis, we compute;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

From our calculator, $n_1 = 5,\ n_2 = 7,\ \bar{x}_1 = 9, \bar{x}_2 = 10, s_1 = 15.5,\ s_2 = 9$.

The pooled standard deviation is given by;

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(5-1)(15.5)^2 + (7-1)9^2}{5 + 7 - 2}} = \sqrt{11.6} = 3.406$$

Therefore;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \frac{9 - 10}{3.406}\sqrt{\frac{5 \times 7}{5 + 7}} = -\frac{1}{3.406} \times 1.708 = -0.5$$

$v = n_1 + n_2 - 2 = 5 + 7 - 2 = 10$. At $\alpha = 5\%$ level of significance,

$t_{\alpha/2}(10) = t_{0.025}(10) = 2.228$

The calculated value of *t* is lessthan the table value.Our null hypothesisis accepted.Hence, we conclude that there isno significantdifference in the efficacy of the twodrugsin the matter of increasingweight.

SPSS OUTPUT

T-TEST GROUPS=Drug(1 2)

/MISSING=ANALYSIS

/VARIABLES=weight

/CRITERIA=CI(.95).

T-Test

Group Statistics

	Drug	N	Mean	Std. Deviation	Std. Error Mean
weigh t	1	5	9.00	3.937	1.761
	2	7	10.00	3.000	1.134

Independent Samples Test

	Levene's Test for Equality of Variances		t-test for Equality of Means						
	F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
								Lower	Upper

weight	Equal variances assumed	.192	.671	-.501	10	.627	-1.000	1.994	-5.444	3.444
	Equal variances not assumed			-.478	7.182	.647	-1.000	2.094	-5.927	3.927

Conclusion

Since computed P-value = 0.627 > 0.05, we accept H_0 and conclude that that there isno significantdifference in the efficacy of the twodrugsin the matter of increasingweight.

Example 4

Two salesmen A and Bare working in a certain district. From a sample survey conducted bythe Head Office,the following results were obtained. State whether there is any significant difference in the average sales between the two salesmen.

	A	B
No of sales	10	18
Average sales (in lakh Rs)	190	205
Standard deviation (in lakh Rs)	20	25

Solution

Null hypothesis; $H_0 : \mu_1 = \mu_2$ *i.e.,* there is no significant difference in the average sales between the two salesmen.

We want to test the hypothesis:

$H_0 : \mu_1 = \mu_2 \quad versus$

$H_0 : \mu_1 \neq \mu_2$

To test the above hypothesis, we compute;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} \text{ where}$$

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(9)(20)^2 + (17)25^2}{10 + 18 - 2}} = \sqrt{\frac{3600 + 10625}{26}} = 23.39$$

Therefore;

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \frac{190 - 205}{23.39}\sqrt{\frac{10 \times 18}{10 + 18}} = \frac{15}{23.39} \times 2.54 = 1.63$$

$v = n_1 + n_2 - 2 = 10 + 18 - 2 = 26$. At $\alpha = 5\%$ level of significance,

$t_{\alpha/2}(26) = t_{0.025}(26) = 2.056$

The table value of t at 5% level ofsignificance for 26 *d.f.* is 2.056.The calculated value of t is less than the table value.The hypothesis holds true. Hence,we conclude that there isno significant difference in the average sales between the two salesmen.

Confidence Interval for the Difference between the Two Means

Two samples of sizes n_1 *and* n_2 are randomly and independently drawn from two normally distributed populations with unknowns but equal variances. The $100(1-\alpha)$ per cent confidence interval for $\mu_1 - \mu_2$ is given by:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}(n_1 + n_2 - 2)\, s / \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

(3) Test of hypothesis about the difference between two means with dependent samples (This is also known as paired t-test)

In the previous section, we assumed that the two random samples drawn from the two populations were independent. In many practical situations, this may not be true. The samples are *dependent* if they are paired so that each observation in one sample is associated with some particular observation in the second sample. Because of this property, the test we are going to use will be called *paired t-test.* In this test, it is necessary that the observations in the two samples be collected in the form called matched pairs. If two samples are dependent, they must have the same number of units. Instead of obtaining two random samples, we can get one random sample of pairs, and the two measurements associated with a pair will be related to each other. This kind of a problem arises in cases such as before and after type experiment or when observations are matched by rise or some other criterion. Suppose that two training methods are to be compared on the basis of average scores by management trainees divided into two equal size classes, one taught by each method. When experimental results are available, we test the null hypothesis that the means associated with the

twomethods are equal, i.e., $H_0 : \mu_1 = \mu_2$. The appropriate test statistic to be used here is

$$t = \frac{\bar{d}\sqrt{n}}{s}$$

follows *t-distribution with (n – 1) d.f.,* where $\bar{d}$ =mean of thedifferences is given by

$\bar{d} = \frac{\sum d}{n}$, s is the standard deviation of the differences and is given by:

$$s = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}} = \sqrt{\frac{\sum d^2 - n(\bar{d})^2}{n-1}}$$

and n is the number of paired observations in the samples.If the computed value of t (at a specified level of significance for a given number of degrees of freedom) is less than the table value of t, our null hypothesis is accepted,otherwise rejected.

Example 5

Ten persons were appointed in an officer cadre in an office. Their performance was noted by giving a test and the marks were recorded out of 100.Theywere given 4 monthstrainingand a test washeld and marks were recorded out of 100.

Employee	A	B	C	D	E	F	G	H	I	J
Before training	80	76	92	60	70	56	74	56	70	56
After training	84	70	96	80	70	52	84	72	72	50

By applying the t-test,can itbe concluded that the employees have benefited bythe training?

Solution

Let us take the null hypothesis that the employeeshave not benefited bythe training. Applying *t-test:*

Employee	Before (1st)	After (2nd)	d=(1st – 2nd)	d2
A	80	84	-4	16
B	76	70	6	36
C	92	96	-4	16
D	60	80	-20	400
E	70	70	0	0
F	56	52	4	16
G	74	84	-10	100
H	56	72	-16	256
I	70	72	-2	4
J	56	50	6	36
			$\sum d = -40$	$\sum d^2 = 880$

$t = \frac{\bar{d}\sqrt{n}}{s}$, where $\bar{d} = \frac{\sum d}{n} = \frac{-40}{10} = -4$,

$s = \sqrt{\frac{\sum d^2 - n(\bar{d})^2}{n-1}} = \sqrt{\frac{880 - 10(-4)^2}{9}} = \sqrt{\frac{880-160}{9}} = 8.944$. Therefore;

$t = \frac{\bar{d}\sqrt{n}}{s} = \frac{-4\sqrt{10}}{8.944} = -1.414.$

$v = n - 1 = 10 - 1 = 9$. At $\alpha = 5\%$ level of significance, $t_{\alpha/2}(9) = t_{0.025}(9) = 2.262$.

The calculated value is less than the table value. Thenull hypothesis holdstrue.Hence,it can beconcluded that the employees have not benefited by the training.

SPSS OUTPUT

T-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	before	69.00	10	12.083	3.821
	after	73.00	10	14.181	4.485

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	before & after	10	.779	.008

Paired Samples Test

	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Pair 1 before - after	-4.000	8.944	2.828	-10.398	2.398	-1.414	9	.191

Conclusion from the SPSS Output

Since computed P-value = 0.191 > 0.05, the null hypothesis is accepted. Hence,it can beconcluded that the employees have not benefited by the training.

Example 6

Ten workers were given a training programme with a view to shorten their assembly time for a certain mechanism. The results of the time and motion studies before and after the training programme are given below:

Worker	1	2	3	4	5	6	7	8	9	10
First study (in mnts)	15	18	20	17	16	14	21	19	13	22
Second study (in mnts)	14	16	21	10	15	18	19	16	14	20

On the basis of this data, can it be concluded that the training programme has shortened the average assembly time?

Solution

Let ustakethenull hypothesisthat the training programme hasnot helped in reducing the average assembly time. Applying the t-test:

$$t = \frac{\bar{d}\sqrt{n}}{s}$$

Calculation for $\bar{d}$ *and* s

Employee	1st study	2nd study	d=(1st – 2nd)	d^2
1	15	14	1	1
2	18	16	2	4
3	20	21	-1	1
4	17	10	7	49

5	16	15	1	1
6	14	18	-4	16
7	21	19	2	4
8	19	16	3	9
9	13	14	-1	1
10	22	20	2	4
			$\sum d = 12$	$\sum d^2 = 90$

$t = \frac{\bar{d}\sqrt{n}}{s}$, where $\bar{d} = \frac{\sum d}{n} = \frac{12}{10} = 1.2$,

$s = \sqrt{\frac{\sum d^2 - n(\bar{d})^2}{n-1}} = \sqrt{\frac{90 - 10(1.2)^2}{9}} = \sqrt{10 - 1.6} = \sqrt{8.4} = 2.898$. Therefore;

$t = \frac{\bar{d}\sqrt{n}}{s} = \frac{1.2\sqrt{10}}{2.898} = 1.309$.

For $v = n - 1 = 10 - 1 = 9$. At $\alpha = 5\%$ level of significance, $t_{\alpha/2}(9) = t_{0.025}(9) = 2.262$

Since the computed value of *t* is less than the table value,the null hypothesis is accepted. Hence,the training programme has not shortened the average assembly time.

SPSS OUTPUT

T-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	First	17.50	10	3.028	.957
	Second	16.30	10	3.302	1.044

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	First & Second	10	.584	.077

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	First - Second	1.200	2.898	.917	-.873	3.273	1.309	9	.223

Conclusion from the SPSS Output

Since computed P-value = 0.223 > 0.05, the null hypothesis is accepted. Hence,the training programme has not shortened the average assembly time.

Confidence Interval for the Mean of the Difference

When the population for the mean of differences is normally distributed with unknown variance for dependent samples, a $100(1-\alpha)$ per cent confidence interval is given by:

$$\bar{d} \pm t_{\alpha/2}(n-1)\, s/\sqrt{n}$$